

8. (a) For a number of years CFCs were used in refrigeration and in air conditioning. The use of these compounds is in decline as they damage the ozone layer. A typical CFC is chlorotrifluoromethane.

Give an equation that shows the formation of a chlorine radical from chlorotrifluoromethane. [1]

- (b) CFCs are being replaced by hydrofluorocarbons (HFCs) such as difluoromethane, which can be made by the reaction of dichloromethane with hydrogen fluoride in the presence of a catalyst.



One problem with this reaction is that chlorofluoromethane is also produced.

The table below shows the percentage of the organic products formed (and unreacted dichloromethane) for various starting ratios of the reactants.

Mole ratio HF : CH ₂ Cl ₂	Percentage composition of mixture / %		
	CH ₂ Cl ₂	CH ₂ ClF	CH ₂ F ₂
26 : 1	1.5	6.5	92.0
22 : 1	1.8	10.2	88.0
19 : 1	3.0	11.0	86.0
16 : 1	4.5	11.5	84.0



- (i) Write a statement that links the mole ratio of $\text{HF} : \text{CH}_2\text{Cl}_2$ with the yield of difluoromethane. [1]

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- (ii) In an experiment using 0.040 mol of CH_2Cl_2 , the yield of CH_2F_2 was 89.0%. Calculate the mass of difluoromethane produced. [2]

Mass = g

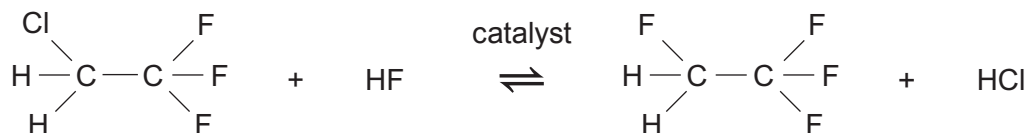


- (c) 1,1,1,2-Tetrafluoroethane is an HFC that is made from trichloroethene and hydrogen fluoride in a two-stage vapour phase process.

Stage 1 – an exothermic process



Stage 2 – an endothermic process



- (i) Suggest, giving a reason, whether a high or low pressure should be used in **stage 1**. [1]

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- (ii) Suggest, giving a reason, whether a high or low temperature should be used in **stage 2**. [1]

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- (iii) The boiling temperatures of the organic products formed after each stage are shown below.

Compound	Boiling temperature / °C
1,1,1-trifluoro-2-chloroethane	7
1,1,1,2-tetrafluoroethane	-26

The final product tends to be contaminated with some of the chlorocompound formed in the first stage.

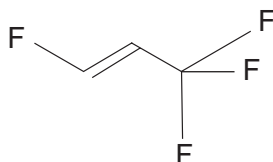
State the name of the method that can be used to separate these two compounds and suggest how this could be carried out in practice. [2]

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- (d) The use of HFCs is now declining as these compounds are contributing to global warming. Compounds such as HFOs (hydrofluoroolefins) are under development as suitable replacements. One such HFO is 1,3,3,3-tetrafluoropropene.



- (i) Explain why 1,3,3,3-tetrafluoropropene exists as *E/Z* isomers. [1]

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- (ii) The addition of hydrogen bromide to 1,3,3,3-tetrafluoropropene results in two structural isomers, each of molecular formula $C_3H_3BrF_4$.

I. State the type of mechanism occurring in this reaction. [1]

- II. These two compounds are formed in **approximately equal amounts** in this reaction.

Suggest why this ratio is different to that found in the reactions of other alkenes. [1]

- (iii) Both the compounds formed in part (ii) will rotate the plane of plane polarised light.

Give the formula of **both** compounds indicating the chiral centre in both. [2]

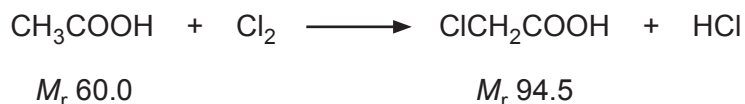
- (iv) Draw the two mirror image forms of the compound 1-bromo-1,2-difluoroethane, $CHBrFCH_2F$. [1]



SECTION B

Answer **all** questions in the spaces provided.

8. (a) Chloroethanoic acid is produced from ethanoic acid by passing chlorine gas into it under suitable conditions.



- (i) This reaction involves radicals. In the first stage chlorine radicals are formed. They then attack the ethanoic acid molecule producing a molecule of hydrogen chloride and a new radical.

Suggest the formula of the new radical formed from the acid during this reaction.

[1]

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- (ii) I. In an experiment 89.0 g of ethanoic acid reacted with chlorine.

Calculate the increase in mass that occurs if chloroethanoic acid is the only organic product. [2]

Increase in mass = g

- II. Chloroethanoic acid reacts with ammonia producing aminoethanoic acid.



In the experiment described in I, 89.0 g of ethanoic acid then gave 49.2 g of aminoethanoic acid (M_r 75.0).

Calculate the percentage yield of aminoethanoic acid. Give your answer to an **appropriate** number of significant figures. [2]

Percentage yield = %



- III. Aminoethanoic acid is isolated from the solution of the products by adding methanol to the liquid. The acid is precipitated as a white solid.

Suggest why aminoethanoic acid is insoluble in methanol. [2]

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- (b) Explain why an aqueous solution of aminoethanoic acid does not affect the plane of plane polarised light. [1]

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- (c) Draw the structure of the dipeptide formed from two molecules of aminoethanoic acid. [1]



- (d) Aminoethanoic acid reacts with nitric(III) acid to produce nitrogen gas.



Calculate the maximum volume of nitrogen gas produced at 100 °C and 98 kPa pressure when 0.300 mol of aminoethanoic acid reacts in this way. Give your answer in dm³. [3]

Volume produced = dm³

- (e) Hydroxyethanoic acid, CH₂(OH)COOH, can also be prepared by reacting chloroethanoic acid with aqueous sodium hydroxide.

Explain why the reaction mixture needs to be acidified to produce hydroxyethanoic acid. [1]

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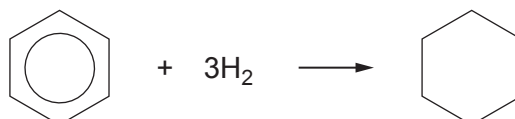
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SECTION BAnswer **all** questions.

8. (a) The starting material for making polyamides is often benzene, obtained from petroleum.
- (i) The first stage in the production of a particular polyamide is the hydrogenation of benzene at increased temperature and pressure.

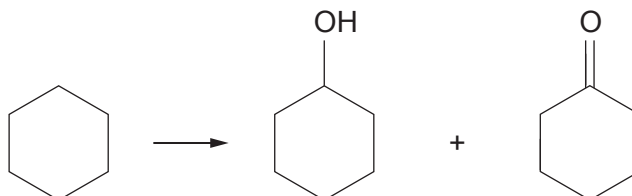


Suggest the name of a catalyst that can be used for the hydrogenation of alkenes and an arene, such as benzene. [1]

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- (ii) Cyclohexane, the product from the first stage, is then catalytically oxidised using air. This results in KA (a mixture of cyclohexanol and cyclohexanone). The mixture typically contains around 60% of cyclohexanone.



- I. Using characteristic infrared values, describe how you could identify both cyclohexanol and cyclohexanone in KA. You should identify the bonds and their absorption frequencies in your answer. [2]

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- II. A student suggested that you could find the proportion of cyclohexanol and cyclohexanone in KA by comparing the percentage of oxygen in each compound.

Calculate the percentage of oxygen in each compound. Use your answers to explain why this method would probably not give an accurate answer for the proportion of each compound present in KA. [3]

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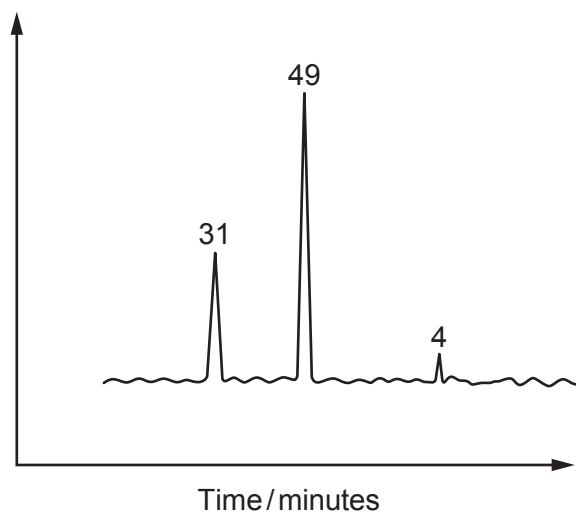
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- III. The proportions present in KA can be found more effectively by using gas chromatography.

A typical chromatogram of a sample of KA is shown below. The largest peak is due to cyclohexanone.



Calculate the percentage by volume of cyclohexanone in the mixture. [1]

Percentage of cyclohexanone = %

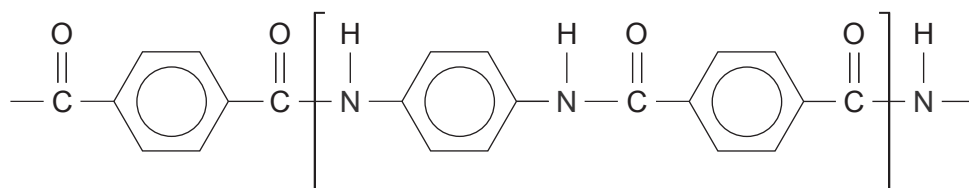
- (iii) A sample of the cyclohexanol in KA is oxidised in the laboratory by refluxing with a suitable reagent.

State a suitable oxidising agent that can be used. [1]

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(b) The formula of the polyamide 'Kevlar' is shown below.

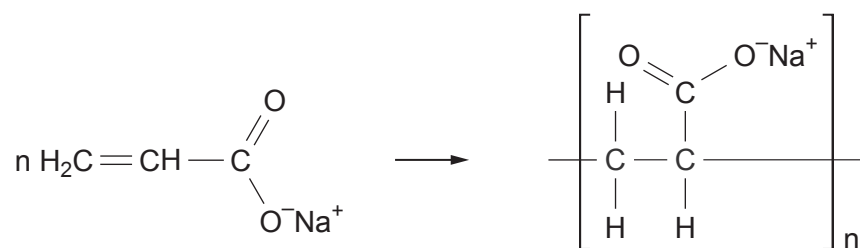


Give the structures of two starting materials that can react together to produce this polymer.

[2]



- (c) The polymer poly(sodiumacrylate) is an example of a super absorbing polymer.



- (i) Explain why this polymer is described as an addition polymer and not as a polyester. [2]

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- (ii) Poly(sodiumacrylate) is capable of absorbing several hundred times its own mass of water, whilst remaining as a solid material.

In an experiment, 4.0×10^{-6} mol of the polymer absorbed 150 g of water.
This mass of water is 300 times the mass of the polymer used.

Calculate the relative molecular mass of the polymer.

[2]

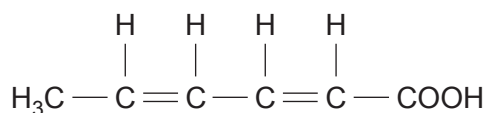
$M_r =$

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Section B

Answer **all** questions.

8. (a) Sorbic acid was isolated in 1859 from mountain ash berry oil. It is a white solid that is slightly soluble in cold water.



- (i) The solubility of sorbic acid in water is 1.6 g dm^{-3} at 20°C and 40.0 g dm^{-3} at 100°C .

Calculate how much sorbic acid is precipitated from its aqueous solution if 200 cm^3 of a saturated solution at 100°C is cooled to 20°C .

Give your answer to an appropriate number of significant figures.

[2]

.....g

- (ii) Sorbic acid and its salts, for example sodium sorbate, have important uses as antimicrobial agents in food preservation.
Some moulds are, however, able to detoxify the action of these sorbates.
An example is the decarboxylation of sodium sorbate.

I. State what is meant by 'decarboxylation'.

[1]

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- II. In the laboratory sodium sorbate can be decarboxylated by heating it with soda lime.

The organic product of decarboxylation is *E*-penta-1,3-diene.

Write the equation for this decarboxylation of sodium sorbate with soda lime (which you should represent as NaOH in your equation), showing the structure of *E*-penta-1,3-diene.

[2]

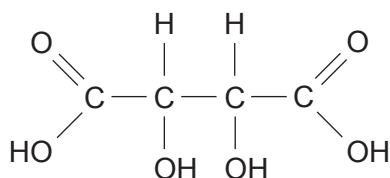


- III. *E*-penta-1,3-diene can then react with hydrogen using platinum as a catalyst giving pentane.

State the number of moles of hydrogen required to react with 0.2 mol of *E*-penta-1,3-diene in this way. [1]

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- (b) Tartaric acid (2,3-dihydroxybutanedioic acid) is used in the food industry.



- (i) Indicate any chiral centre(s) present by means of an asterisk (*). [1]

- (ii) This acid occurs in a number of optically active forms.
Complete the sentences below.

Forms of tartaric acid that rotate the plane of plane-polarised light are called

..... [1]

A solution containing an equimolar mixture of two forms of the acid that rotates the plane of plane-polarised light in equal and opposite directions is called a

..... [1]

- (c) Both sorbic acid and tartaric acid are described as showing stereoisomerism.

Explain what is meant by the term stereoisomerism. [1]

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- (d) Tartaric acid is formed when butenedioic acid reacts with a suitable oxidising agent.

Give the equation for this reaction, representing the oxidising agent as [O] and using water as a reactant. [1]



- (e) The solubility of unsaturated carboxylic acids in water decreases as the chain length increases.

Number of carbon atoms in the alkyl chain	Solubility in water at 25 °C / g dm ⁻³
3	94
5	1.6
7	0.7

Suggest why this solubility decreases as the chain length increases.

[2]

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